

Neel Maddu

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S./M.S. Computer Science | **GPA: 4.0**

B.S. Dec 2027 | M.S. Dec 2028

- **Coursework:** Machine Learning, Artificial Intelligence, Systems & Networks, Automata & Complexity, Objects & Design, Data Structures & Algorithms, Applied Combinatorics.
- **U.S. Citizen** | Eligible for security clearance.

EXPERIENCE

Georgia Tech Research Institute (GTRI)

May 2026 – Present

Software Engineer Intern, CIPHER Lab – Threat Analysis Branch

Atlanta, GA

- Build NLP pipelines for entity extraction and semantic analysis across **60K+** unstructured text records and posts, supporting cybersecurity and intelligence analysis workflows in a national security context.
- Evaluate **15 agentic configurations** against **6 categories** of adversarial and injection-style attacks across an isolated **3-VM** testbed, quantifying how prompt design, tool permissions, and guardrail placement drive failure rates.
- Brief **5 program stakeholders** on model-vulnerability findings, informing security assessments and program-level architecture decisions.

Georgia Institute of Technology – VIP Program

Jan. 2026 – Present

Automated Algorithm Design – AlphaEvolve Subteam

Atlanta, GA

- Extend the LLM-Guided Evolution (LLM-GE) framework – cited as prior work by Google DeepMind’s AlphaEvolve – running and validating **7 generations** of C880 circuit-optimization experiments on Georgia Tech’s PACE ICE HPC cluster.
- Contribute to a GTRI-sponsored multi-objective genetic programming framework that evolves hybrid algorithms from data, outperforming human-designed baselines.

Georgia Tech iOS Club

Aug. 2025 – Present

Tech Lead

Atlanta, GA

- Co-lead a **20-member** project team alongside another Tech Lead and 4 Senior Developers, architecting and shipping production iOS applications in Swift, SwiftUI, and Firebase.
- Own MVVM feature architecture and run code review across the team codebase, onboarding new developers and enforcing mobile development standards each semester.

PROJECTS

MARTA Live Transit Tracker | Python, FastAPI, SQLite, SwiftUI, WidgetKit, GTFS-Realtime

2026

- Built a full-stack transit tracker: a 24/7 FastAPI/SQLite backend polling MARTA’s GTFS-Realtime and rail feeds, accumulating **575K+ arrival observations** across 78 routes and 6,200+ stops, feeding a native SwiftUI app tracking ~200 live vehicles at 15-second refresh.
- Resolved feed defects absent from official docs – reconstructed missing bus delays via trip-to-schedule joins against **2.4M GTFS rows** and hand-wrote a protobuf decoder validated byte-for-byte against the reference parser – and integrated OpenTripPlanner for delay-aware trip planning, backed by **59 automated tests**.

LLM Red-Teaming Evaluation Pipeline | Python, Streamlit, Ollama, Anthropic API, pytest

2026

- Built an extensible prompt-injection evaluation framework (pluggable provider/target/classifier/defense layers) testing local and frontier LLMs against **46 attack payloads** across 4 categories and 41 techniques, with 35 tests and CI.
- Engineered a dual rule-based + LLM-as-judge scoring pipeline and Streamlit dashboard quantifying a robustness gap of up to **47% breakthrough** on local 8B models versus near-zero on a frontier model.
- Surfaced non-obvious failure modes via multi-model, repeated-trial experiments – prompt-hardening defenses that relocated vulnerabilities rather than removing them, and denylist filters defeated by trivial obfuscation.

Multimodal AI Search Engine | PyTorch, FAISS, NumPy, pandas, Streamlit

2025

- Indexed **10,000+ images** with CLIP embeddings and FAISS, retrieving semantically relevant results from natural language prompts at an average query time of 2 seconds.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, Swift, SQL

ML & AI: PyTorch, TensorFlow, scikit-learn, FAISS, MediaPipe, NLP, LLM Evaluation, Graph Analytics

Frameworks & Libraries: FastAPI, Node.js, React, Next.js, SwiftUI, Core ML, Streamlit, pandas, NumPy

Data & Infrastructure: MySQL, MongoDB, SQLite, Firebase, Docker, Git, GitHub Actions, Jenkins, CI/CD